

David Xia

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EDUCATION

• University of Illinois Urbana-Champaign

August 2023 - May 2027

Bachelor of Science in Computer Science

Champaign, Illinois

Bachelor of Science in Liberal Arts & Sciences Major in Mathematics, Data Optimization Concentration

Bachelor of Science in Liberal Arts & Sciences Major in Statistics

◦ GPA: 3.98/4.00

PAPERS

SUB=SUBMITTED, ACC=ACCEPTED

[1] [SUB] Revitalizing Local Democracy: A Human-Centered Audit of LLMs in City Council Journalism [arXiv]

David Xia, Chris Maury

Submitted to CHI 2026 HEAL Workshop

[2] [SUB] RSK linear operators and the Vershik-Kerov-Logan-Shepp curve [arXiv]

Duy Phan, David Xia

Submitted to Electronic Journal of Combinatorics

[3] [ACC] EveGuard: Defeating Vibration-based Side-Channel Eavesdropping with Audio Adversarial Perturbations [arXiv]

Jung-Woo Chang, Ke Sun, David Xia, Xinyu Zhang, Farinaz Koushanfar

IEEE Symposium on Security and Privacy, 2025

RESEARCH EXPERIENCE

• Carnegie Mellon University Human-Computer Interaction Institute , Research Intern

May 2025 - present

- Designed and evaluated an automated journalism pipeline using SOTA LLMs to replicate end-to-end editorial workflows, from transcript segmentation to headline generation and topic prioritization.
- Orchestrated large-scale crowd-sourced study to establish human-evaluated ground truth validation.
- Demonstrated that LLM-based headline quality and topic prioritization can meet and exceed professional-standards.
- Presented research poster at the CMU HCII REU Symposium.
- Paper in submission to CHI 2026 Human-centered Evaluation and Auditing of Language Models (HEAL) workshop.

• Illinois Mathematics Lab (formerly Illinois Geometry Lab) , Student Scholar

Jan 2024 - present

- Researched the data science perspective of sports matches at the Illinois Mathematics Lab.
- Built web scrapers with Selenium to scrape online data and analyzed the predictability of sports matches by comparing prediction accuracies of multiple sources including polls and the betting market.
- Evaluated efficiency of probabilistic predictions based on the betting market, Elo, and other mathematical models.
- Presented findings at Rose-Hulman Undergraduate Math Conference 2024, UIUC Undergraduate Research Symposium 2024, and Joint Mathematics Meeting 2025.
- Currently working on paper.

• Illinois Combinatorics Lab for Undergraduate Experience (ICLUE) , Research Mentee

Aug 2024 - May 2025

- Studying algebraic combinatorics, focusing on the Robinson-Schsted-Knuth correspondence.
- Used the Vershik-Kerov-Logan-Shepp curve to apply probability bounds on the RSK linear operator.
- Paper in submission to the Electronic Journal of Combinatorics.

• National University of Singapore SEEDER Group , Research Intern

May 2024 - Jul 2024

- Developed machine learning pipeline to detect, classify, and analyze diseases found in plant leaves.
- Trained U-Net segmentation models and classification models based on convolutional neural networks to isolate and classify images using the TensorFlow framework, achieving 90% segmented classification rate.
- Generated synthetic images using custom trained cGAN model to increase data variety and volume.
- Stacked deep learning classification models with traditional machine learning models to achieve an increased detection accuracy of over 95%.

• University of California San Diego, Research Intern

Jun 2020 - Aug 2023

- Developed EveGuard, a software-based defense framework to protect voice privacy from vibrometry-based side channels using adversarial audio by using a perturbation generator model (PGM) to effectively suppress sensor-based eavesdropping while preserving high audio quality.
- Implemented Eve-GAN, a novel domain translation task for inferring eavesdropped signals, enabling end-to-end training of PGM, utilizing few-shot learning techniques to reduce data collection overhead.
- Achieved a protection rate of over 97% against audio classifiers, hindering eavesdropped reconstruction.
- Paper published in the IEEE Symposium on Security and Privacy 2025.

PRESENTATIONS

- 2025 **Carnegie Mellon University Human-Computer Interaction Institute REU Symposium** LLMs for Extracting Agenda Items from City Council Meetings and Generating Newsworthy Headlines
- 2025 **Algebra, Geometry and Combinatorics Day** RSK linear operators and the Vershik-Kerov-Logan-Shepp curve (joint work presented by Duy Phan)
- 2025 **Joint Mathematics Meeting Pi Mu Epsilon Poster Session** Predictability in Major Sports Leagues
- 2024 **Illinois Mathematics Lab Research Symposium** Predictability in College Sports
- 2024 **University of Illinois Undergraduate Research Symposium** Predictability in College Sports
- 2024 **Rose-Hulman Undergraduate Math Conference** Predictability in College Sports

SERVICE

- SP26 **University of Illinois Urbana-Champaign** Algorithms and Models of Computation (CS 374) Course Assistant

GRANTS AND AWARDS

- 2025 **CMU HCII REU: NSF 2349558**
- 2025 **ICLUE RTG: NSF 1937241**